CONTROLLER / REV A HARDWARE REVIEW

One LKS32MC451VCT8; 2 inverter ports + 1 dedicated PFC port. No LLC, USB or Hall.
All sheets share the explicitly named global nets. All external ports are NON-ISOLATED in this draft.

5 V input, 3.3 V and supervision

5 V input -> TLV76733 -> 3.3 V
TPS3808G33 reset and MCU decoupling

文件: power.kicad_sch

LKS32MC451VCT8 signal assignment

U1A/B/C + crystal + reserved test pads
Physical pin numbers follow the 100-pin CSV

文件: mcu.kicad_sch

Inverter 1 signal port and brake

J2: six PWM + enable + fault + eight ADC
J3: separate brake command

文件: inverter_1.kicad_sch

Inverter 2 signal port and brake

J4: six PWM + enable + fault + eight ADC
J5: separate brake command

文件: inverter_2.kicad_sch

Dedicated Boost / Vienna control port

J6: six gate commands + nine ADC
Enable, fault and future precharge command

文件: pfc.kicad_sch

ABZ encoder ports

J7/J8: independent A/B/Z inputs
3.3 V single-ended draft; optional power DNP

文件: sensors.kicad_sch

Two independent SPI expansion ports

J9/J10: independent SPI_A / SPI_B
CS pull-ups; optional sensor power DNP

文件: spi.kicad_sch

SWD, UART, CAN and status LEDs

J11 SWD / J12 UART / J13 CAN option
Reset, five status LEDs; CAN defaults DNP

文件: debug_comms.kicad_sch

REVIEW BOUNDARIES
1. Only a regulated 5 V source enters J1; select one inverter supply. No parallel supplies.
2. Three 24-pin ports carry signals only; no bus power or sensor supply on these ports.
3. GND/AGND contact positions share controller GND. Different power reference domains require isolation.
4. ABZ is provisionally 3.3 V single-ended. Do not connect 5 V / differential encoders directly.
5. Connector mechanics, crystal loading, ESD/clamps and component procurement remain review items.
6. PFC timing / ADC scheduling are not proven by this schematic. See design_review.md.

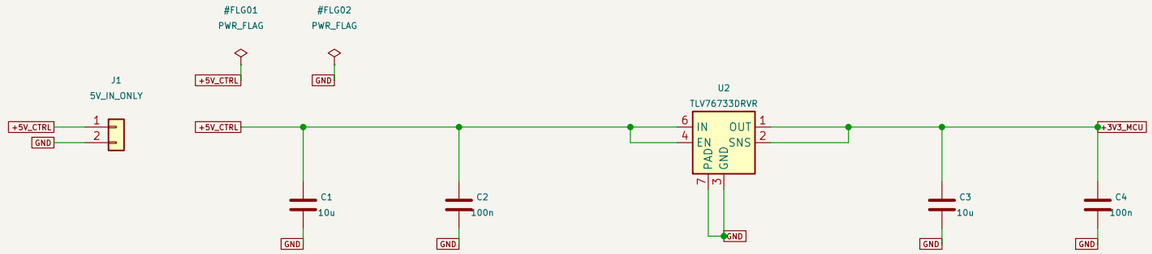
LKS32MC451VCT8 | single reference GND | 5 V input only
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /
File: controller.kicad_sch

Title: Controller — Overview and design boundaries

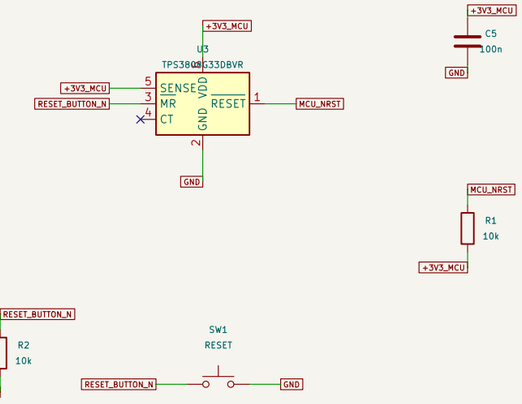
Size: A3	Date: 2026-11-01	**Rev: A-draft**
KiCad E.D.A. 10.0.7		Id: 1/9

5 V AUXILIARY INPUT –> 3.3 V



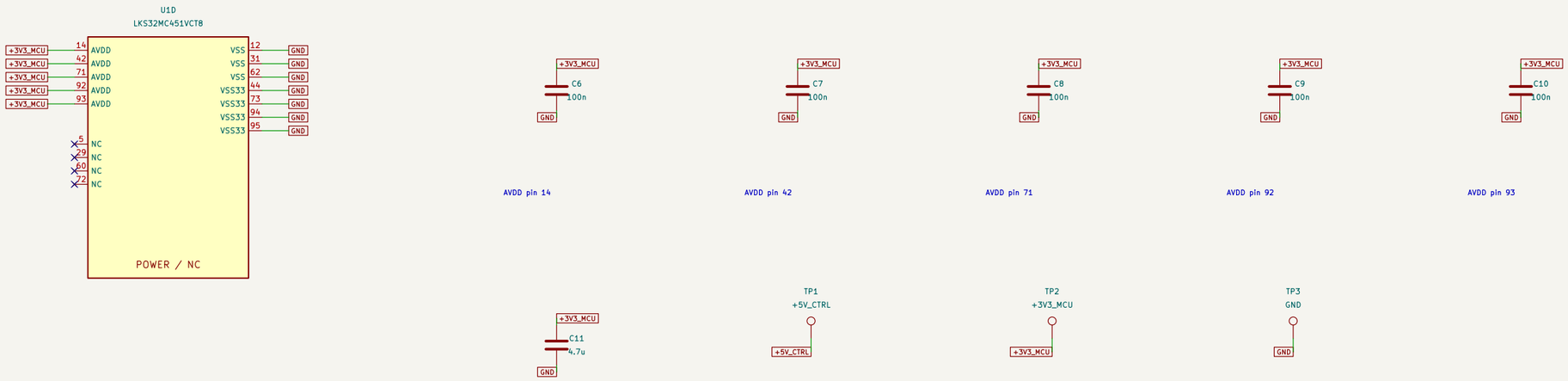
U2: EN follows input; SNS tied to OUT. All GND pins and EP go to GND.
CIN/COUT: 10 uF X7R/X5R, >=16 V, 0805; verify effective C >=1 uF.
Budget: 250 mA –> (5–3.3)*0.25 = 0.425 W. Copper/EP thermal review required.
5 V input protection/current limit is on the designated inverter (TPS2553 branch).

POWER MONITOR / MCU RESET



G33 nominal falling threshold 3.07 V. CT intentionally open: 20 ms nominal release.
SW1 drives MR; SWD NRST also directly pulls MCU_NRST low.
RESET is open-drain, with 10k pull-up. No large reset capacitor on the SWD line.

MCU POWER UNIT / LOCAL DECOUPLING



All five AVDD pins connect to the same 3.3 V rail. Continuous ground plane; no split return slit.
Each 100 nF at its own AVDD/VSS pair. NC 5/29/60/72 remain unconnected.
REF (74) and LDO12 (B5) are multiplexed test outputs, not supply inputs.

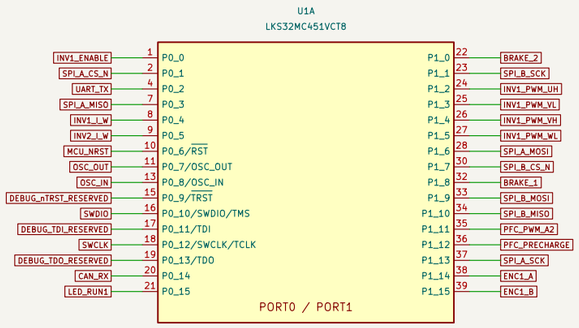
LKS32MC451VCT8 | single reference GND | 5 V input only
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

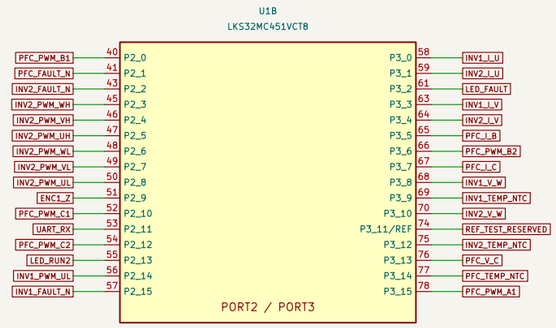
Sheet: /5 V input, 3.3 V and supervision/
File: power.kicad_sch

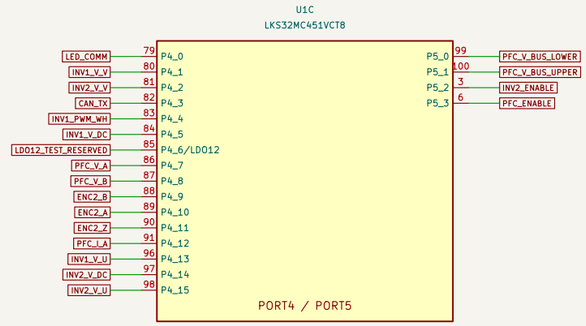
Title: Controller — 5 V input, 3.3 V and supervision

Size: A3	Date: 2026-11-01	**Rev: A-draft**
KiCad E.D.A. 10.0.7		Id: 2/9

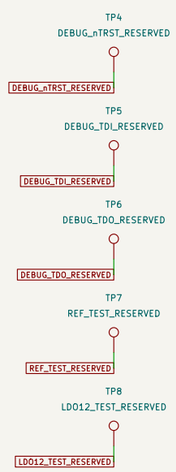
EXACT PHYSICAL PIN ASSIGNMENT / THREE GPIO UNITS



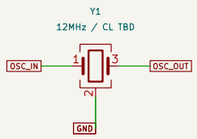




RESERVED DEBUG / ANALOG TEST OUTPUTS



EXTERNAL CRYSTAL



DS v1.71 section 8: oscillator amplifier and load capacitors are internal.
No external load capacitors populated. Select crystal CL/ESR after vendor confirmation.
Keep both crystal traces short, symmetric, and away from gate/PWM routing.

Keep SWDMUX=0, RST_IO=0 and TRACEMUX=0 for this allocation.
Reserved test outputs are high-impedance pads only; no attached load or capacitor.
ADC pin mapping and selected AF remain in LKS32MC451VCT8_pin_plan.csv.
No board identification circuit consumes the reserved connector pins.

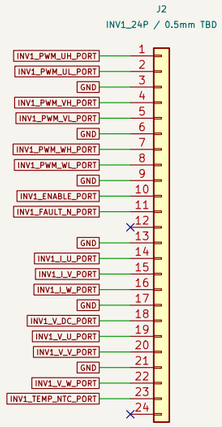
LKS32MC451VCT8 | single reference GND | 5 V input only
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /LKS32MC451VCT8 signal assignment/
File: mcu.kicad_sch

Title: Controller — LKS32MC451VCT8 signal assignment

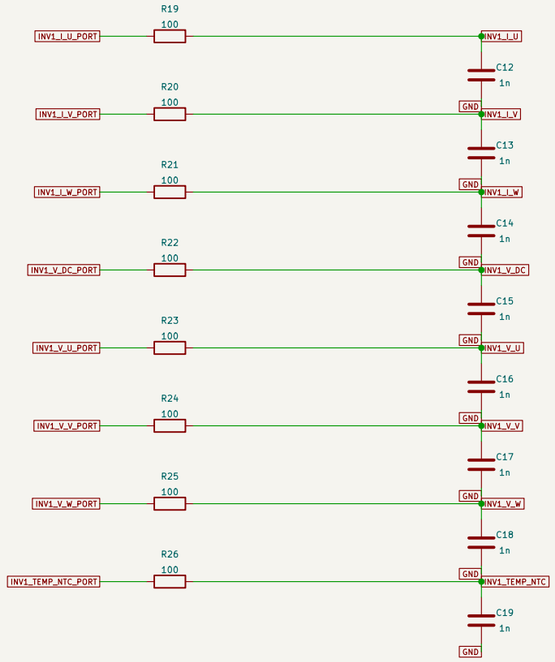
Size: A3	Date: 2026-11-01	**Rev: A-draft**
KiCad E.D.A. 10.0.7		Id: 3/9

INV1 / 24-PIN SIGNAL CONNECTOR



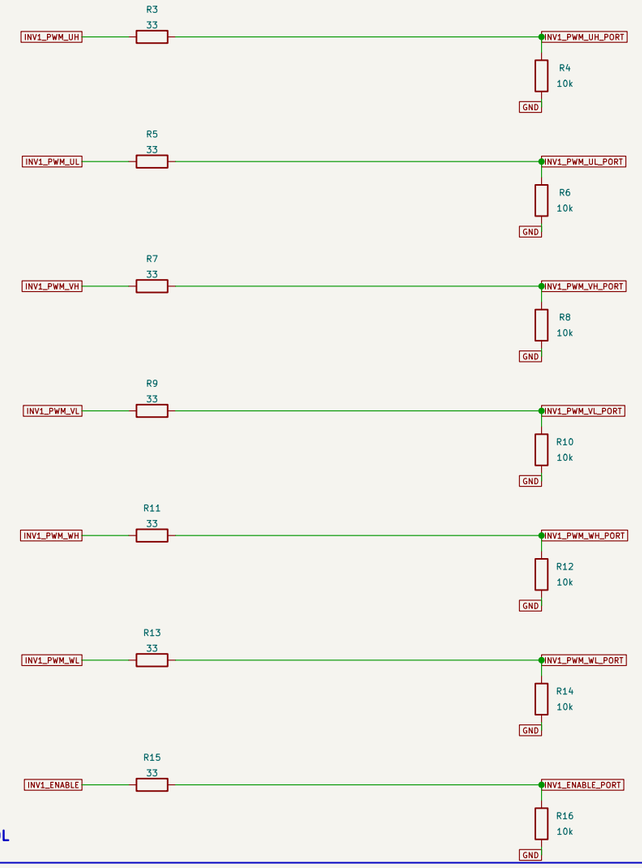
Contacts 3/6/9/13/17/21 -> controller GND. AGND is a return name, not isolation.
No auxiliary power on this connector. Footprint, latch and cable contact orientation TBD.
Use same-reference laboratory boards only until the isolation architecture is reviewed.

ADC INPUTS / PROVISIONAL RC

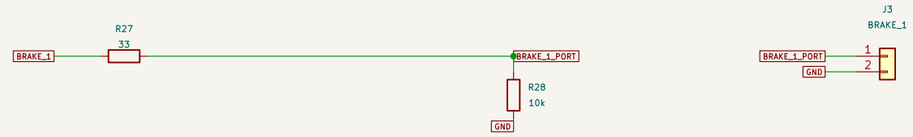


I: 1.65 V +/-50 mV/A; V: 50:1, NTC scale TBD.
Inputs must remain within 0..3.3 V while controller is powered.
100R/1nF: tau=100 ns, fc=1.59 MHz; validate source + ADC settling.
No overvoltage / power-off backfeed protection claimed by this RC.

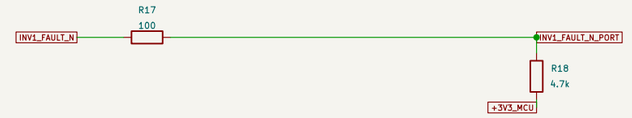
DIGITAL COMMANDS / RESET DEFAULT LOW



SEPARATE BRAKE CONTROL



FAULT INPUT / ACTIVE LOW OPEN-DRAIN



No RC delay in fast fault path. Local power board protection acts first.
FAULT pull-up alone does not detect an unplugged cable.

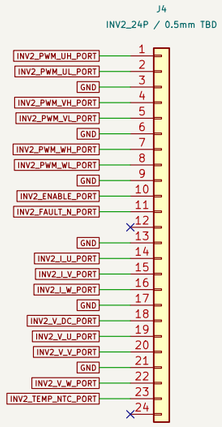
LKS32MC451VCT8 | single reference GND | 5 V input only
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Inverter 1 signal port and brake/
File: inverter_1.kicad_sch

Title: Controller — Inverter 1 signal port and brake

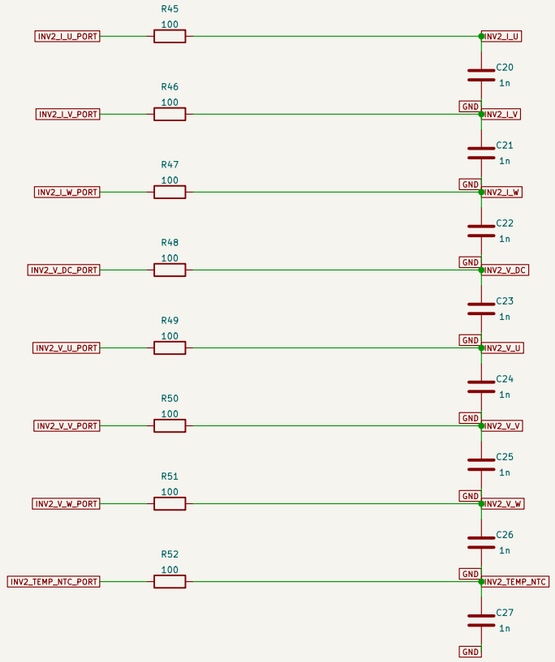
Size: A3	Date: 2026-11-01	**Rev: A-draft**
KiCad E.D.A. 10.0.7		Id: 4/9

INV2 / 24-PIN SIGNAL CONNECTOR



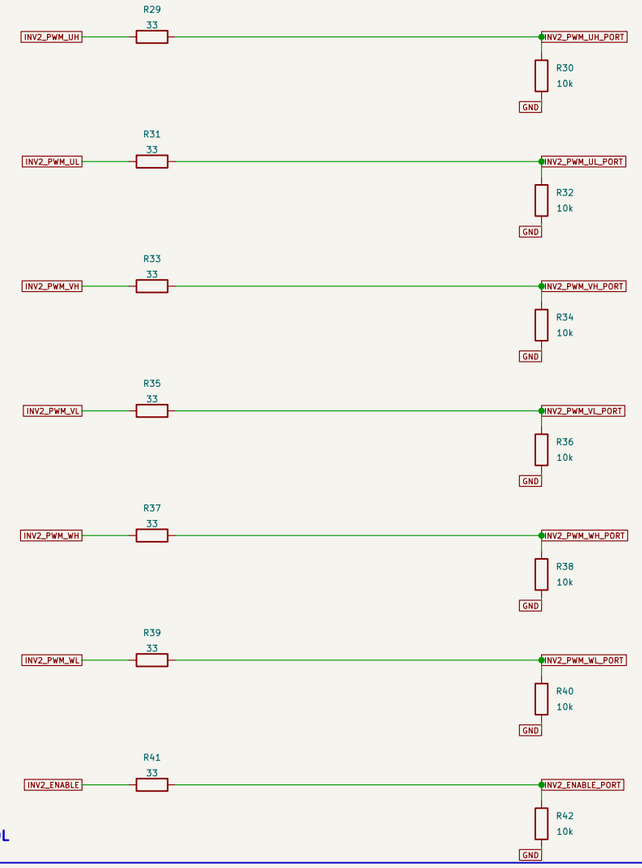
Contacts 3/6/9/13/17/21 -> controller GND. AGND is a return name, not isolation.
No auxiliary power on this connector. Footprint, latch and cable contact orientation TBD.
Use same-reference laboratory boards only until the isolation architecture is reviewed.

ADC INPUTS / PROVISIONAL RC

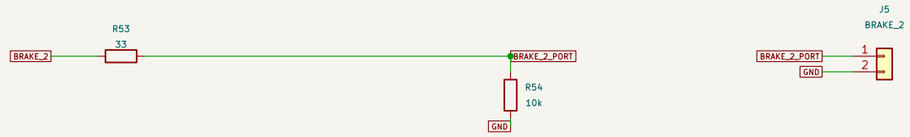


I: 1.65 V +/-50 mV/A; V: 50:1, NTC scale TBD.
Inputs must remain within 0..3.3 V while controller is powered.
100R/1nF: tau=100 ns, fc=1.59 MHz; validate source + ADC settling.
No overvoltage / power-off backfeed protection claimed by this RC.

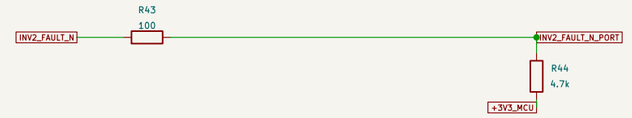
DIGITAL COMMANDS / RESET DEFAULT LOW



SEPARATE BRAKE CONTROL



FAULT INPUT / ACTIVE LOW OPEN-DRAIN



No RC delay in fast fault path. Local power board protection acts first.
FAULT pull-up alone does not detect an unplugged cable.

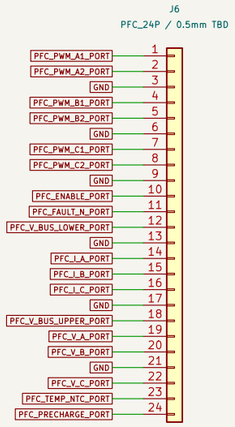
LKS32MC451VCT8 | single reference GND | 5 V input only
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Inverter 2 signal port and brake/
File: inverter_2.kicad_sch

Title: Controller — Inverter 2 signal port and brake

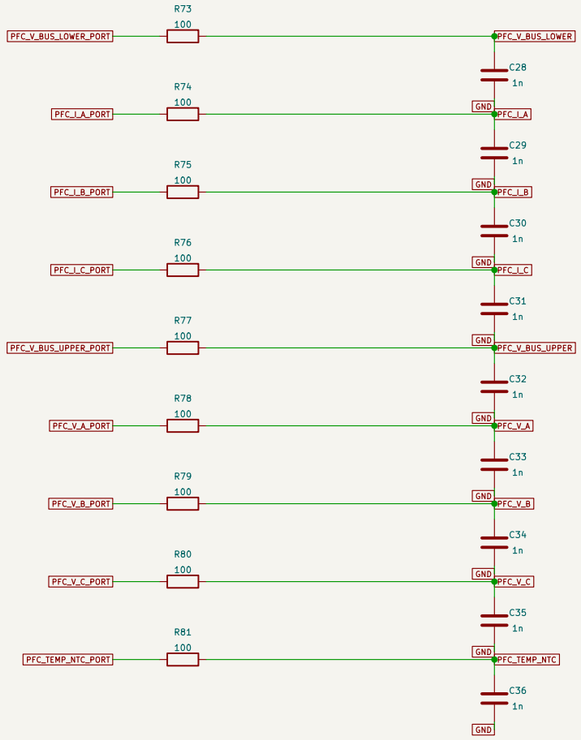
Size: A3	Date: 2026-11-01	**Rev: A-draft**
KiCad E.D.A. 10.0.7		Id: 5/9

PFC / 24-PIN SIGNAL CONNECTOR



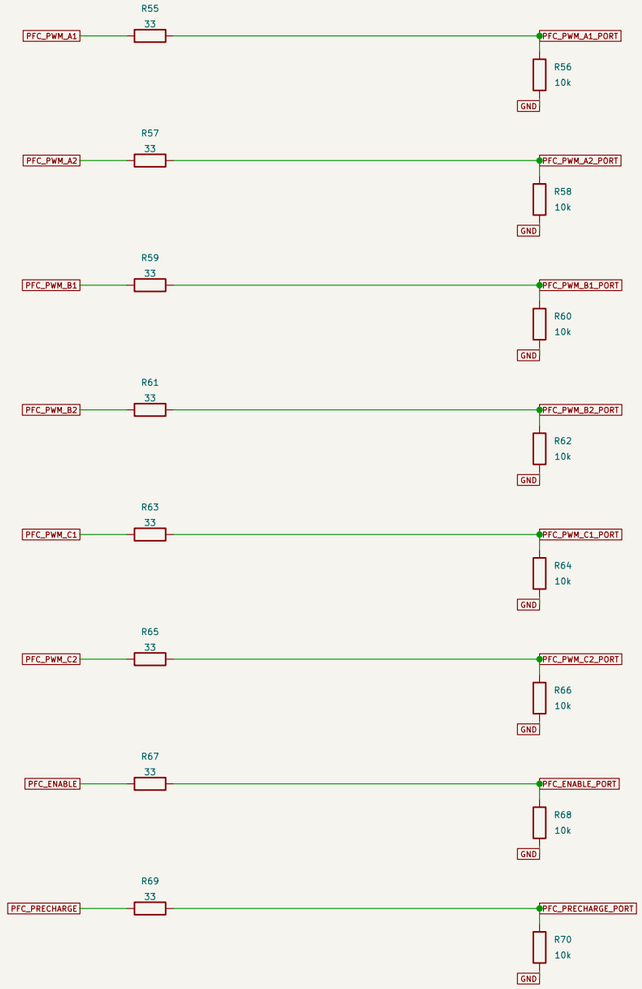
Contacts 3/6/9/13/17/21 -> controller GND. AGND is a return name, not isolation.
No auxiliary power on this connector. Footprint, latch and cable contact orientation TBD.
Use same-reference laboratory boards only until the isolation architecture is reviewed.

ADC INPUTS / PROVISIONAL RC

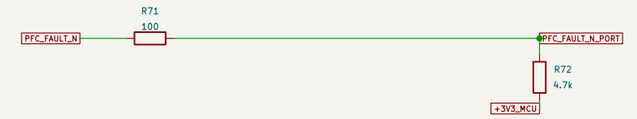


Boost uses A1: Vienna gate pairing / timing remains a validation task.
PFC PRECHARGE remains reserved: inverter precharge was cancelled.
ADC 100R/1nF: tau=100 ns. PFC scaling / NTC / source impedance TBD.
ADC input clamps, unpowered pin injection and cross-domain isolation TBD.

DIGITAL COMMANDS / RESET DEFAULT LOW



FAULT INPUT / ACTIVE LOW OPEN-DRAIN



No RC delay in fast fault path. Local power board protection acts first.
FAULT pull-up alone does not detect an unplugged cable.

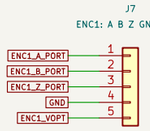
LKS32MC451VCT8 | single reference GND | 5 V input only
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

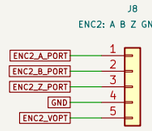
Sheet: /Dedicated Boost / Vienna control port/
File: pfc.kicad_sch

Title: Controller — Dedicated Boost / Vienna control port

Size: A3	Date: 2026-11-01	**Rev: A-draft**
KiCad E.D.A. 10.0.7		Id: 6/9

ABZ INCREMENTAL ENCODERS / 3.3 V SINGLE-ENDED DRAFT





































VOPT link + bypass DNP: no sensor power by default.
3.3 V push-pull ABZ only; no differential receiver or 5 V conversion.
External supply reference must be GND; do not backfeed this port.

VOPT link + bypass DNP: no sensor power by default.
3.3 V push-pull ABZ only; no differential receiver or 5 V conversion.
External supply reference must be GND; do not backfeed this port.

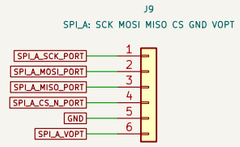
ENCODER INTERFACE REVIEW
The five-pin connector is a provisional schematic assignment, not a frozen production connector.
Actual encoder model / output stage / cable length / pulse rate must be selected before PCB.
A 5 V TTL, open-collector, RS-422 or other differential encoder needs an appropriate input stage.
The eight reserved sensor-power parts on the ABZ/SPI sheets are not enabled by merely fitting connectors.

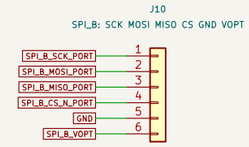
LKS32MC451VCT8 | single reference GND | 5 V input only
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system
Sheet: /ABZ encoder ports/
File: sensors.kicad_sch

Title: Controller — ABZ encoder ports

Size: A3	Date: 2026-11-01	**Rev: A-draft**
KiCad E.D.A. 10.0.7		Id: 7/9

TWO INDEPENDENT SPI MASTERS / 3.3 V



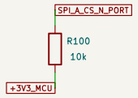














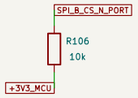
















VOPT: optional 3.3 V power link and bypass are DNP. No sensor power is supplied by default.
CS_N has 10k pull-up; both SPI peripherals remain unselected during MCU reset.
Peripheral current budget, ESD, power sequencing, cable rate/termination and final pin order remain open.
Short internal cables only; do not connect 5 V logic or drive unpowered controller pins.

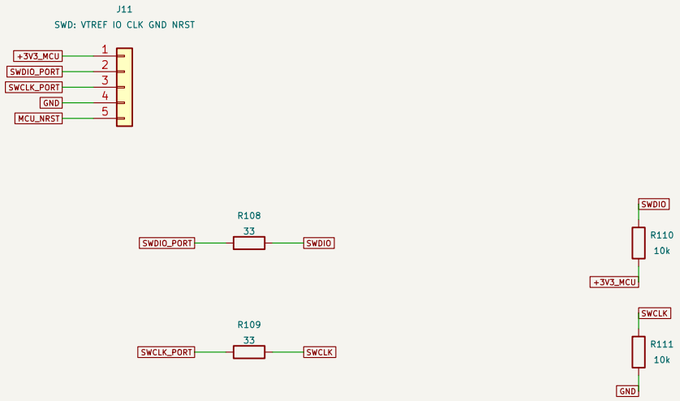
LKS32MC451VCT8 | single reference GND | 5 V input only
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Two independent SPI expansion ports/
File: spi.kicad_sch

Title: Controller — Two independent SPI expansion ports

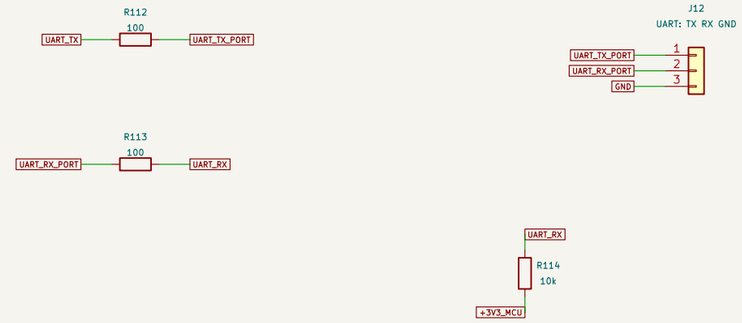
Size: A3	Date: 2026-11-01	**Rev: A-draft**
KiCad E.D.A. 10.0.7		Id: 8/9

SWD / RESET



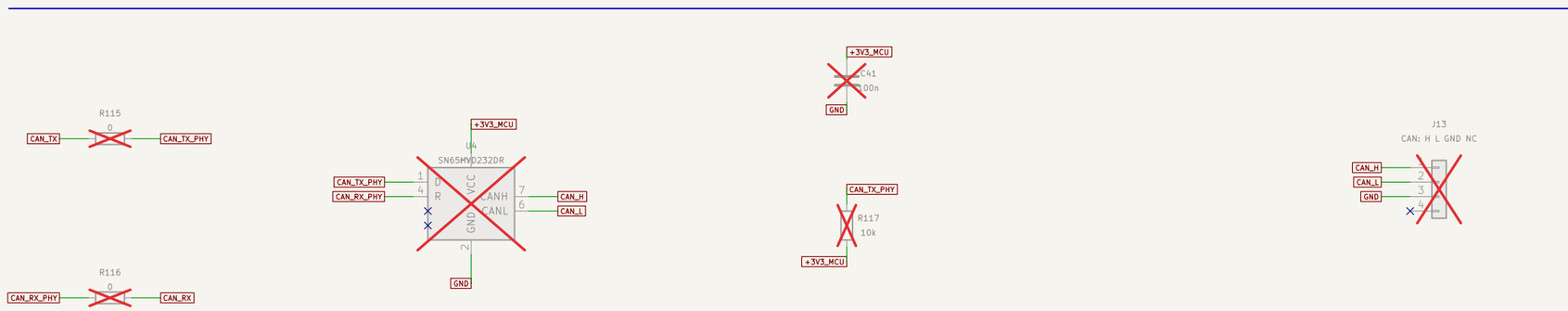
VTREF is target-voltage reference, not a power input.
No SWO; preserve reserved JTAG group pins.
MCU_NRST open-drain pull-up is on power sheet.

3.3 V TTL UART / NO POWER PIN



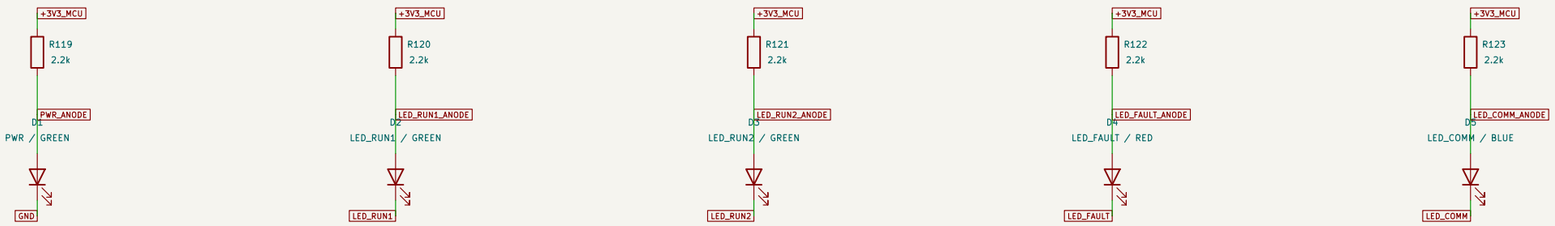
TX/RX directions relative to controller. Cross TX to peer RX.
3.3 V logic only: no RS-232 / RS-485 transceiver.

CLASSIC CAN / OPTIONAL ASSEMBLY BLOCK



DNP block: U4, two OR links, TX pull-up, bypass, J and termination.
Populate the entire transceiver block together; 120R only at a bus end.
SN65HVD232 pins 5/8 are NC; no standby GPIO.
Pin 4 at connector is NC; no supply output. ESD/protection TBD.

FIVE INDICATORS / LOW-ACTIVE GPIO LEDS



Indicative LED current <=(3.3-1.6)/2.2k=0.77 mA. LEDs do not grant hardware PWM permission.
Exact LEDs and switch are procurement TODOs.

LKS32MC451VCT8 | single reference GND | 5 V input only
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /SWD, UART, CAN and status LEDs/
File: debug_comms.kicad_sch

Title: Controller — SWD, UART, CAN and status LEDs

Size: A3	Date: 2026-11-01	**Rev: A-draft**
KiCad E.D.A. 10.0.7		Id: 9/9